Uncertainty, samples and cause

How sure can we be of a number from a sample, and when does a pattern mean that one thing caused another?

AI and Data Science Professional | Session 4 of 24 | Module A: Think with data
KLAi Professional Learning | Jotham Martin Wambi

Today in one view: what you will leave with

By the end of today you can



Explain why a sample differs from the population, and measure that difference with the standard error



Run a Monte Carlo simulation with RAND in Google Sheets and read the sampling distribution it produces



Read a Pearson correlation, say what it does not show, and name a confounder and two kinds of bias

Your two hours: concepts first, then a Google Sheets lab

0:00	10	Recall and the question of the day
0:10	45	Concepts: picture, plain words, one example by hand
0:55	10	Break
1:05	45	Lab: simulate, correlate, question
1:50	10	Exit check, keep and next

Ideas with pictures first. Then you open the sheet. Then the checks and what you keep.

Three questions from last time

1. A manager asks what a typical order is worth. Which statistic do you lead with, and what else do you give?

2. What is a pivot table in SQL and in pandas?

3. A bar chart makes a small gap between two branches look huge. What do you check first?



Answer aloud first. Then we check together.

Answers to the three questions from Session 3

	Question	Short answer
	**A manager asks what a typical order is worth. Which statistic do you lead with, and what else do you give?**	The median, 24,800 UGX, with the number of orders and the IQR. The mean, 49,931, is pulled up by a few large orders.
	**What is a pivot table in SQL and in pandas?**	A GROUP BY in SQL and groupby in pandas: group the records, then aggregate each group.
	**A bar chart makes a small gap between two branches look huge. What do you check first?**	Whether the vertical axis starts at zero. Cutting the base makes small gaps look large.

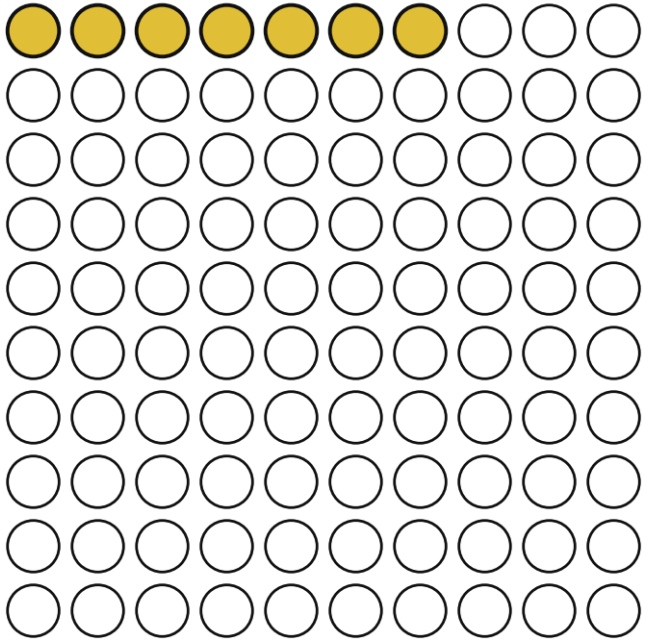
Words we use today

	Word	What it means
	**Population and sample**	The population is every record you care about. A sample is the part you actually look at. The gap between a sample statistic and the population value is sampling error. How much a statistic changes from sample to sample is sampling variability.
	**Standard error**	The standard deviation of a statistic across many samples, such as the sample mean. It says how far one sample can sit from the truth.
	**Monte Carlo simulation**	Many random draws done by computer to see what a statistic does. In Sheets: RAND and INDEX, filled down.
	**Correlation (Pearson r)**	A number from -1 to +1 for how closely two numeric variables rise and fall together. Correlation is not causation.
	**Confounder**	A variable that influences both of two variables you are comparing, so they look linked. Also called a confounding variable.

Probability is a share between 0 and 1: about 7 orders in 100 are above 100,000 UGX

7%

139 of 2,000 orders are above 100,000 UGX. P(order above 100,000) = 139 ÷ 2,000 = 0.0695.



orders in every 100 are above 100,000 UGX

A share settles as you count more records: this is the long run

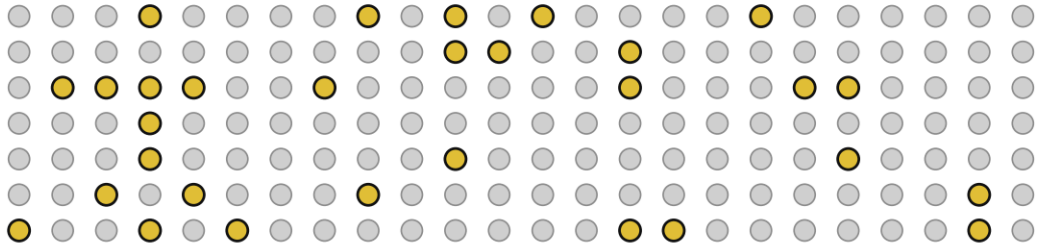


Early shares swing. Later shares settle. This is the law of large numbers.

A sample is drawn from a population, and a statistic from the sample estimates a parameter

Population: 2,000 orders

For this lesson, the orders in the workbook. A number that describes the whole population is a parameter: the population mean is 49,931 UGX.



random sample, n = 30

Sample: 30 orders

Drawn at random. A number computed from the sample is a statistic: here the sample mean is 33,396 UGX.

Sampling error = sample mean minus population mean = 33,396 minus 49,931 = -16,535 UGX

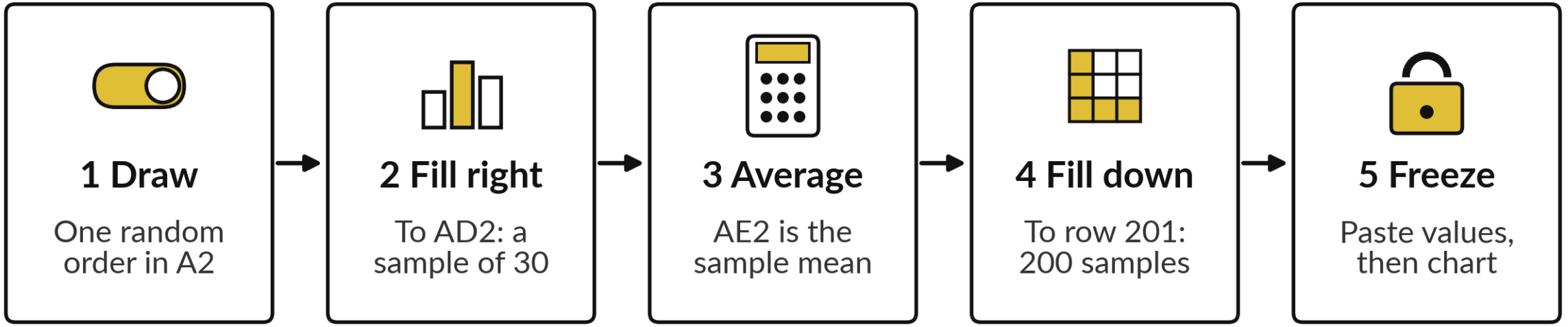
The workbook is itself a sample of the full file. Here we treat its 2,000 orders as the population.

Three random samples of 30 orders give three different means, and the gaps are sampling error

	Sample	Sample mean (UGX)	Sampling error (UGX)
	Sample 1	33,396	-16,535
	Sample 2	36,610	-13,321
	Sample 3	34,348	-15,583
	Population (all 2,000 orders)	49,931	0

These samples come from one seeded run of the simulation. Yours will differ. Sampling error = sample mean minus population mean.

A Monte Carlo simulation repeats a random draw many times: five steps in Sheets



Each draw puts a random row number into INDEX. Drawing the same row twice is allowed: sampling with replacement.

Three small formulas build one random draw and one sample mean

Piece	Formula	What you get
A random decimal	=RAND()	A number from 0 up to just below 1, new on every recalculation
A random row number	=1+INT(RAND()*2000)	A whole number from 1 to 2000
One random order	=INDEX(orders!$G$2:$G$2001,1+INT(RAND()*2000))	The order value in that row
The sample mean	=AVERAGE(A2:AD2)	One mean for the row of 30 orders

=RANDBETWEEN(1,2000) is a shortcut for the second formula. Both can give the same row twice.

RAND recalculates on every edit, so freeze the results before you chart them

Live formulas

Every edit redraws all 6,000 random draws and 200 means.

Your histogram changes while you look at it.

Sheets has no setting to fix the draw.

Frozen values

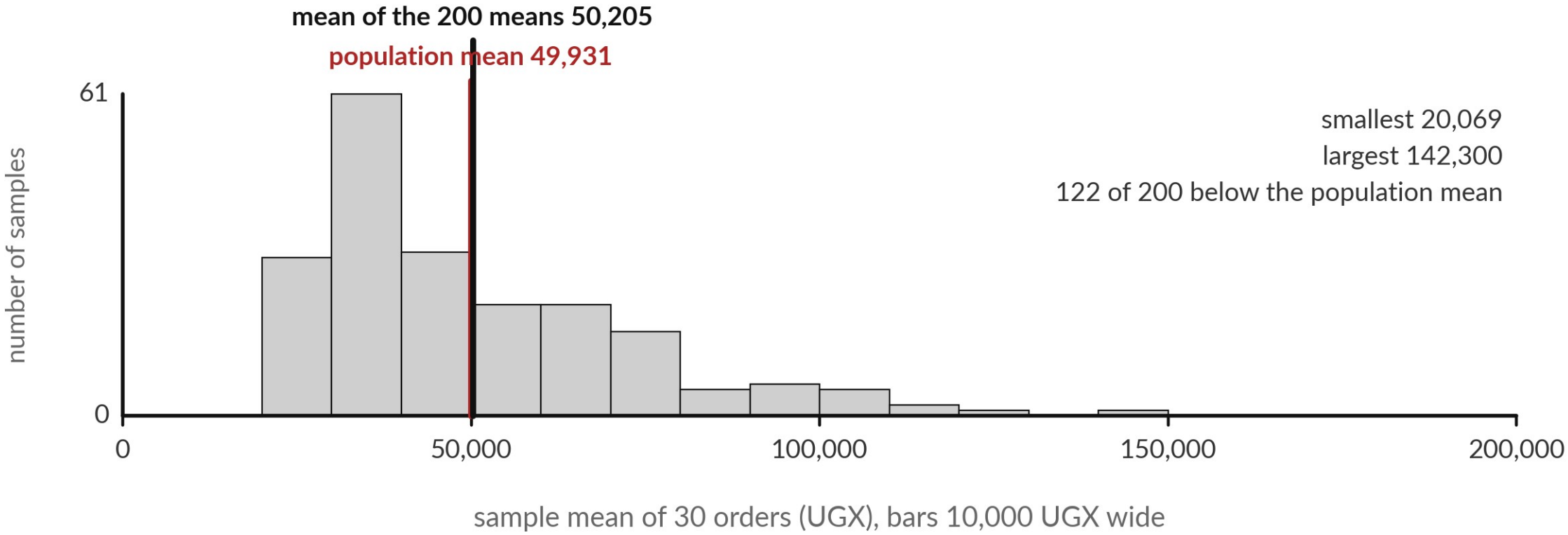
Copy the block, then Paste special, Values only.

The numbers stay fixed.

Now compute and chart.

Sheets cannot fix a draw, so freeze the values.

200 samples of 30 orders give 200 different means: together they form the sampling distribution



One seeded run of the exercise. Your own run will look similar but not identical.

The standard error is the standard deviation of the sample means

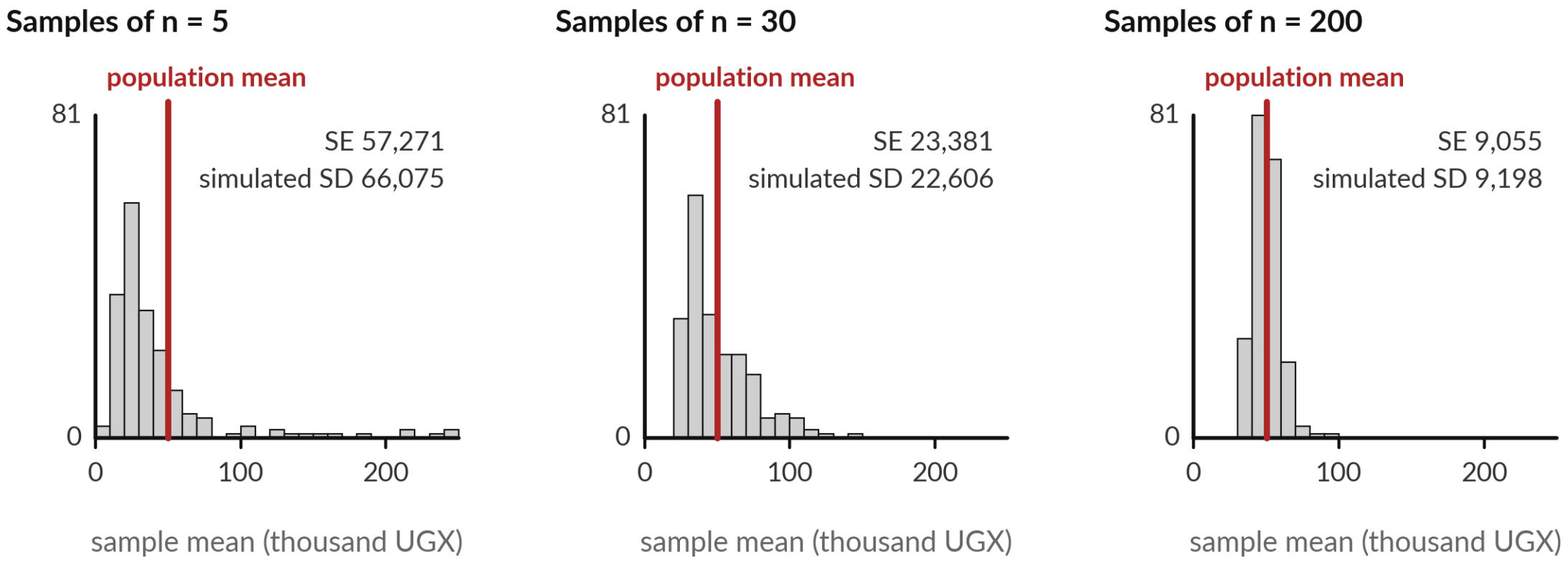
Standard error of the mean

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}} = \frac{128{,}062}{\sqrt{30}} = 23{,}381$$

Symbol	What it means	In our data
SE	Standard error: the spread of the sample means across many samples	Simulated: 22,606 (STDEV of 200 means)
sigma	Population standard deviation (STDEV.P)	128,062 UGX
n	The sample size	30 orders
x-bar	The sample mean	One value in each sample row

With one real sample you do not know sigma, so you use s, the sample standard deviation.

A larger sample has a smaller standard error: quadruple n to halve it



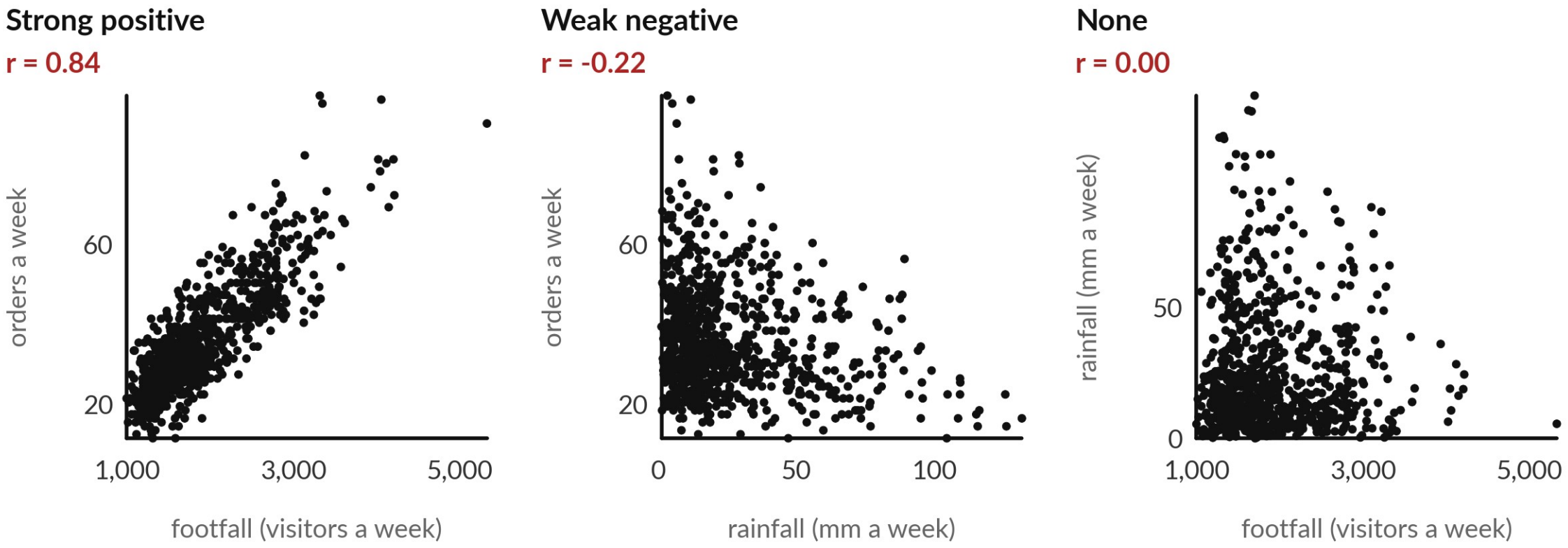
Each panel shows 200 sample means on the same scale. SE = sigma divided by the square root of n.

Sample means of skewed data stay skewed at small n and become bell-shaped as n grows

	Sample size n	Standard error	Skewness of the sample means	Share of samples below the population mean
	5	57,271	3.95	76 percent
	30	23,381	1.63	65 percent
	200	9,055	0.64	54 percent

From 20,000 repeats of each sample size. Order values have skewness 8.76. This pattern is the central limit theorem at work.

Correlation (Pearson r) measures how closely two variables rise and fall together



Each dot is one branch in one week: 832 records from the branch_weeks tab.

Pearson r multiplies how far two variables sit from their means, record by record

Pearson correlation

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} = 0.84$$

Symbol	What it means	In our data
r	Pearson correlation coefficient, from -1 to +1	0.84 for footfall and orders
x_i, y_i	Footfall and orders of branch-week i	832 records
x-bar, y-bar	The mean footfall and the mean orders	Means of the two variables
r squared	Share of the variance in orders that a straight line on footfall accounts for (coefficient of determination)	0.71

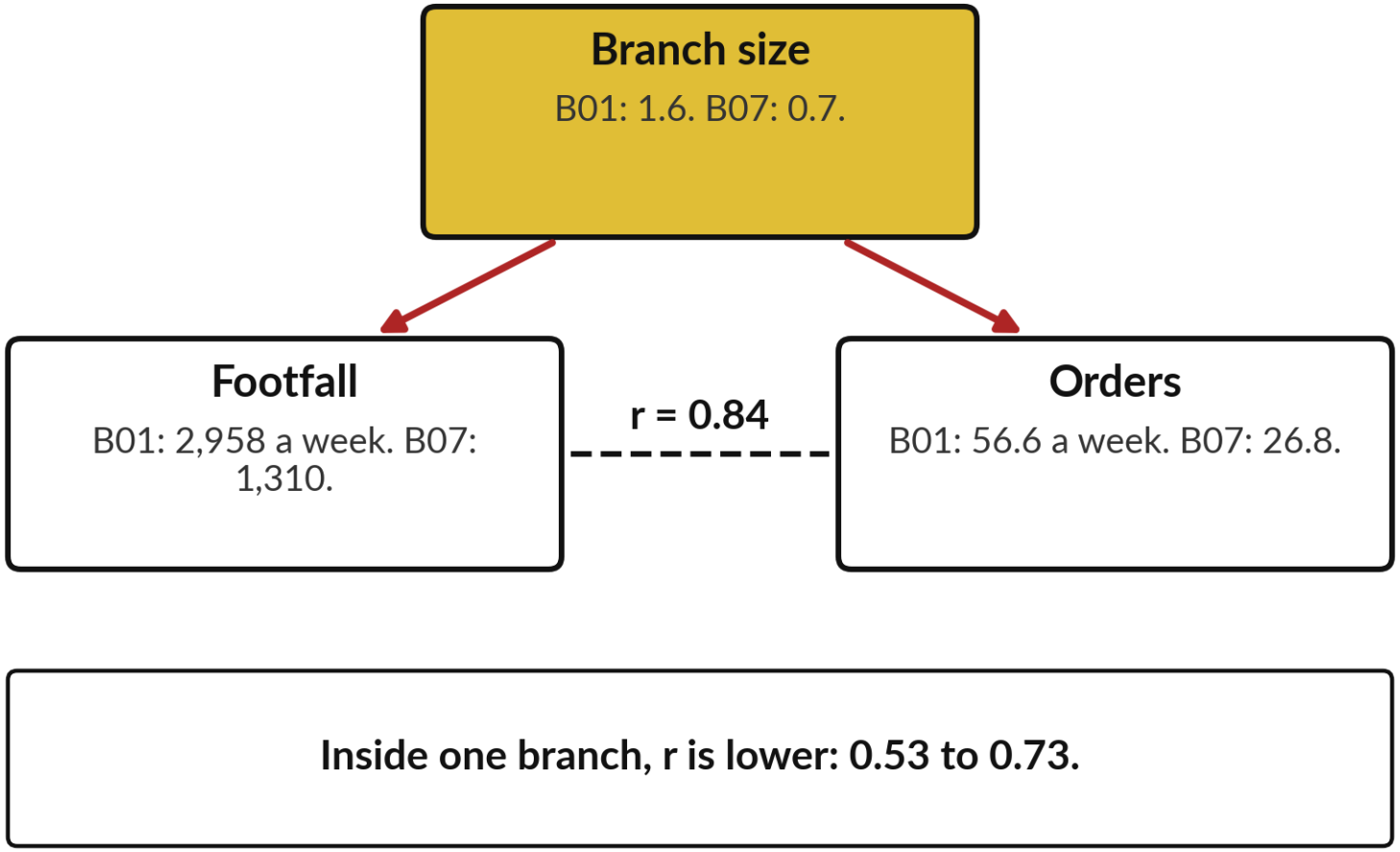
In Sheets: =CORREL(branch_weeks!C2:C833,branch_weeks!F2:F833). Pearson r measures only straight-line association.

Reading r: size, sign and what it cannot tell you

	Pair of variables	r	Reading
	**Footfall and orders**	0.84	Strong, positive. Busier weeks have more orders. It says nothing about which causes which.
	**Rainfall and orders**	-0.22	Weak, negative. Wetter weeks have somewhat fewer orders, with a lot of scatter.
	**Footfall and rainfall**	0.00	About zero. No straight-line link.
	**Malaria cases and rain, same month, District Clinics**	0.00	Near zero, yet with last month's rain r is 0.37. A delay can hide a real link.

Cut-offs such as strong and weak are judgement, not law. Always draw the scatter plot before you trust r.

A confounder influences both variables, so they look linked



Confounder (confounding variable): a variable that drives both of the variables you compare.

Larger branches have more footfall and more orders.

Part of the 0.84 comes from branch size.

Correlation alone cannot show cause.

Bigger branches have more footfall and more orders, so size confounds the correlation

Branch	Size (relative)	Mean footfall a week	Mean orders a week
B01 Kampala Central	1.6	2,958	56.6
B02 Kampala Ntinda	1.3	2,433	46.4
B03 Entebbe	0.9	1,683	32.2
B04 Jinja	1.0	1,852	34.2
B05 Mbale	0.8	1,505	29.2
B06 Mbarara	1.0	1,914	37.0
B07 Fort Portal	0.7	1,310	26.8
B08 Gulu	0.8	1,489	28.2

Across the 8 branches, size and mean footfall have r = 0.999, and size and mean orders have r = 0.996.

Selection bias and survivorship bias both come from records that are missing



Selection bias

Records not chosen by chance.

Workbook means: loyalty 48,672, 580 walk-ins 53,013.

Full file: 47,432 and 47,628.



Survivorship bias

Studying only what is still there.

Hypothetical: judging a strategy only from branches that are still open.

Ask who or what could not get into the sample.

Where you will use this at work: say how far a figure can sit from the truth, and what else could explain it

Your job	The question	What today lets you do
Credit analyst auditing a loan book, using SQL	We checked a sample of loans. How far can the sample default rate sit from the whole book?	Report the standard error with the sample figure.
Monitoring and evaluation officer running a household survey	Is a sample of 300 households enough?	Estimate the standard error, and say who is missing from the sample.
Sales analyst at a retail chain, using Excel	Busy shops sell more. Do we fund flyers?	Name a confounder, such as branch size, before you spend.
Health records officer at a district office, using pandas	Do rainy months bring malaria cases?	Pair each month's malaria_cases with last month's rainfall_mm in the same district.

Read the recipe before the lab: five formulas run the whole lab

```
=INDEX(orders!$G$2:$G$2001,1+INT(RAND()*2000))
=AVERAGE(A2:AD2)
=STDEV(AE2:AE201)
=COUNTIF(AE2:AE201,"<"&AVERAGE(orders!G2:G2001))
=CORREL(branch_weeks!C2:C833,branch_weeks!F2:F833)
```

In plain words

Line 1 draws one random order. Fill it right to AD for one sample of 30.

Line 2 gives the sample mean of that row. Fill down to row 201.

Line 3 is the standard error estimate from your 200 means: about 19,185 to 27,760.

Line 4 counts how many sample means fall below the population mean: about 114 to 146 of 200.

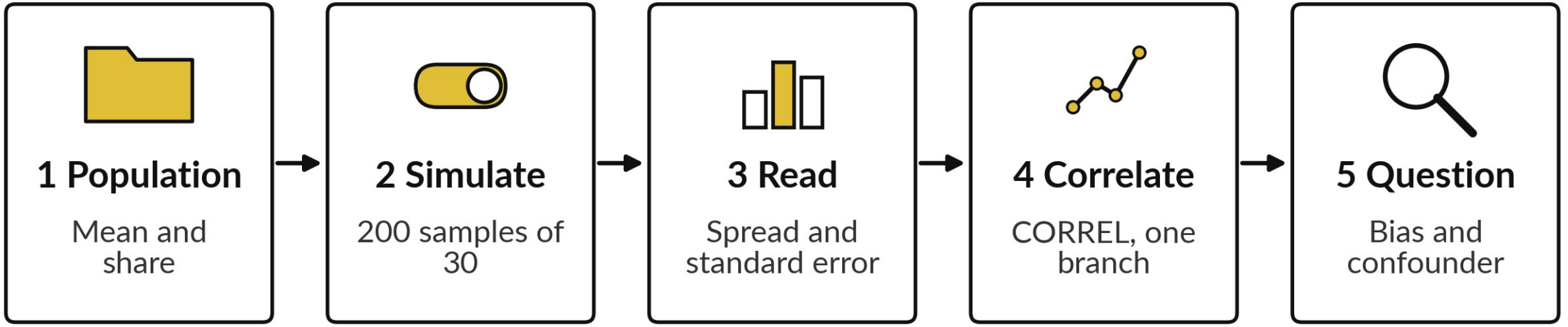
Line 5 gives r for footfall and orders: 0.84.

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Part 2: the Sheets lab

Close the slides. Open the sheet.

The lab has five steps, and you keep a simulation tab at the end



Eight tasks in 45 minutes. For each task, predict the answer, compute it, then write it on your simulation tab.

Check your numbers: fixed answers and ranges for your random ones

	What you compute	You should get
🖩	**Mean of the orders, and orders above 100,000**	49,930.68 and 139
🖩	**Formula =STDEV.P(orders!G2:G2001)/SQRT(30)**	23,381
📊	**Smallest and largest sample mean**	16,293 to 24,427, and 112,051 to 211,224
📊	**STDEV of the 200 means (n = 30)**	19,185 to 27,760
📊	**Sample means below the population mean**	114 to 146 of 200
📊	**STDEV for n = 5 over n = 30**	1.6 to 3.3
📈	**r for footfall and orders: all weeks, then B01 only**	0.84 and 0.70
🔒	**Mean order of loyalty orders and of walk-ins**	48,672 and 53,013

Ranges cover the middle 98 percent of 5,000 repeats of this lab. If you fall outside, look for a range that is too short.

Five likely mistakes in the lab and how to fix them

What you see	What it means	What to do
Your numbers change whenever you edit a cell	The block is still live RAND formulas.	Copy the block, then Paste special, Values only.
#REF! or #NAME? in the INDEX formula	The tab name is missing or misspelt.	Write orders!$G$2:$G$2001 with the tab name and dollar signs.
All 200 means are the same	The block was pasted as values after one draw.	Fill the formula again, then freeze the whole block.
Standard deviation near 128,000	You took STDEV of the orders, not of the 200 means.	Point STDEV at the AE column, AE2:AE201.
CORREL gives #N/A	The two ranges have different lengths.	Use C2:C833 and F2:F833, both 832 rows.

Exit check: three questions, then compare with the short answers

	Question	Short answer
	**Samples of 30 give a standard error of 23,381 UGX. What happens to it with samples of 120?**	It halves, to about 11,690, because it is sigma divided by the square root of n.
	**A manager sees r = 0.84 between footfall and orders and wants to fund flyers. What do you say?**	Branch size is a confounder: it lifts both. Inside one branch r is lower. Test flyers at some branches against similar branches without.
	**Why is reporting only loyalty-card orders risky?**	It is selection bias: the records were chosen by something other than chance, and walk-in shoppers are left out.

Save your simulation tab, then bring one question to Session 5

What you keep from today

Keep your simulation tab: 200 samples of 30 orders, their histogram, and three sentences on what a correlation does and does not show.

Our question today

How sure can we be of a number from a sample, and when does a pattern mean that one thing caused another?

Next session

Session 5: your first Python: variables, lists, dictionaries and error messages. The pandas DataFrame comes in Session 7.

Take away

Your lab file

A simulation tab

Your finding

Three sentences on cause

Next session

Your first Python